

Forward/Backward Sweep (FBS)

Step-by-Step Guide, with Two Worked Examples to Verify Your Code Against

A plain-language method for solving power flow on a **radial** distribution network (a tree with one substation/slack bus and no loops). The steps below work for 5, 33, or any number of buses — nothing changes except how many times you repeat the loop.

1. What You Need Before You Start

Data	What it is
Base values	S_{base} (MVA), V_{base} (kV) — used to convert everything to per-unit (p.u.)
Bus list	Which bus is the slack (substation), and its fixed voltage (usually $1.0 \angle 0^\circ$ p.u.)
Line list	For each line: From-bus, To-bus, R (p.u.), X (p.u.)
Loads	P and Q (kW/kVAr, converted to p.u. by dividing by S_{base}) at each bus
PV / DG units	Treat as a negative load: subtract the PV's P (and Q) from that bus's load

Per-unit conversion: $Z_{base} = V_{base}^2 / S_{base}$. If impedances are already in p.u., skip this; if given in Ohms, divide by Z_{base} . For loads: $P_{pu} = P_{kW} / (S_{base,MVA} \times 1000)$.

2. Draw the Tree and Number the Levels

Starting at the slack bus (“Level 0”), trace outward. Every bus reached after one line is “Level 1”, after two lines is “Level 2”, and so on. Each bus except the slack has **exactly one** line feeding it from upstream — that is what makes the network radial. Write down, for every bus, who its *parent* (upstream bus) is, and who its *children* (downstream buses) are.

Example tree used in the worked examples below:

Bus1 (slack) $\rightarrow$ Bus2 $\rightarrow$ Bus3 $\rightarrow$ Bus4 $\rightarrow$ Bus5

3. Flat Start

Set every bus voltage to the slack voltage (e.g., $V = 1.0 \angle 0^\circ$ p.u.) as your initial guess. Only the slack bus keeps a **fixed** voltage throughout — it never changes.

4. Backward Sweep — Compute Currents, Leaves $\rightarrow$ Root

Work from the outermost buses (leaves) inward toward the slack. For each bus b :

1. Compute the current its own load draws, using its current voltage estimate:

$$I_{load}(b) = \text{conj}(S(b) / V(b))$$

where $S(b) = P(b) + jQ(b)$ is the net load at that bus (load minus PV, if any), and $\text{conj}()$ means take the complex conjugate.

2. The current flowing into bus b on its upstream line equals its own load current plus the sum of the currents already found on every line leaving b toward its children:

$$I(\text{parent} \rightarrow b) = I_{\text{load}}(b) + \sum I(b \rightarrow \text{child})$$

Because you process leaves first, every child's current is already known by the time you need it for its parent.

5. Forward Sweep — Update Voltages, Root → Leaves

Now walk back out from the slack toward the leaves. For each bus b (parent before child):

$$V(b) = V(\text{parent}) - I(\text{parent} \rightarrow b) \times Z(\text{parent} \rightarrow b)$$

where Z is the line's impedance ($R + jX$) in p.u. This updates every bus voltage using the currents just computed.

6. Check Convergence

Compare the new voltages to the previous round's. If the largest change in any bus voltage magnitude is below your tolerance (commonly $1e-6$ p.u.), **stop — you're done**. Otherwise, return to Step 4 (backward sweep again, using the newly updated voltages — this slightly changes the load currents, since I_{load} depends on V).

7. Once Converged — Get Your Final Answers

- **Bus voltages:** the $V(b)$ values from the last forward sweep (magnitude + angle).
- **Branch currents:** the $I(\text{parent} \rightarrow b)$ values from the last backward sweep.
- **Branch power flow** (entering the line at its sending end): $S_{\text{send}} = V(\text{from}) \times \text{conj}(I(\text{parent} \rightarrow b))$
- **Line losses:** $S_{\text{loss}} = |I|^2 \times Z$ (real part = kW lost, imaginary part = kVAr lost). Add up all branches for total system loss.

Worked Example 1 — Tiny 3-Bus Network

Do this one by hand to build intuition before trusting any code with a bigger network.

Setup

All per-unit (base assumed = 1, so p.u. numbers are the actual values):

- Bus 1 = slack, $V = 1.0 \angle 0^\circ$
- Line 1→2: $Z = 0.01 + j0.02$ • Line 2→3: $Z = 0.01 + j0.02$
- Load at Bus 2: $S = 0.30 + j0.15$ • Load at Bus 3: $S = 0.20 + j0.10$

Iteration 1

Backward: $I(2 \rightarrow 3) = I_{\text{load}}(3) = \text{conj}(0.20 + j0.10 / 1.0) = \mathbf{0.2000 - j0.1000}$ ($|I| = 0.2236$)

$I_{\text{load}}(2) = \text{conj}(0.30 + j0.15 / 1.0) = 0.3000 - j0.1500$

$I(1 \rightarrow 2) = I_{\text{load}}(2) + I(2 \rightarrow 3) = \mathbf{0.5000 - j0.2500}$ ($|I| = 0.5590$)

Forward: $V_2 = 1.0 - (0.5 - j0.25)(0.01 + j0.02) = \mathbf{0.99000 - j0.00750}$ ($|V_2| = 0.99003$)

$V_3 = V_2 - (0.2 - j0.1)(0.01 + j0.02) = \mathbf{0.98600 - j0.01050}$ ($|V_3| = 0.98606$)

Max voltage change from flat start = 0.0175 → not converged, repeat.

Iteration 2 (recompute load currents using the new voltages)

$I(2 \rightarrow 3) = 0.201737 - j0.103568$, $I(1 \rightarrow 2) = 0.503602 - j0.257370$

$V_2 = 0.989817 - j0.007498$ ($|V_2| = 0.989845$)

$V_3 = 0.985728 - j0.010497$ ($|V_3| = 0.985784$)

Max change = 0.00027 → still repeat (shrinking fast).

Converges on Iteration 4 (change < $1e-6$)

Bus	V (p.u.)	angle (deg)
1	1.000000	0.000
2	0.989843	-0.4341

3	0.985780	-0.6103
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Branch	$ I $ (p.u.)	$S_{\text{send}} (P + jQ)$	Loss $(P + jQ)$
1→2	0.565684	0.503714 + j0.257429	0.003200 + j0.006400
2→3	0.226832	0.200515 + j0.101029	0.000515 + j0.001029

Total loss = 0.003715 + j0.007429 p.u. Confirm your code reproduces these exact numbers before trusting it on a bigger network.

Worked Example 2 — 5-Bus Radial Network with PV

(The same network as in your Excel workbook — use this to check a larger case.)

Setup

Base 1 MVA / 12.66 kV. Tree: Bus1(slack)→Bus2→Bus3→Bus4 (has 180 kW PV), and Bus2→Bus5.
 Loads (kW/kVAr): Bus2 = 250/120, Bus3 = 300/150, Bus4 = 200/100 (net of PV: 20/100), Bus5 = 150/80.
 Lines (p.u.): 1-2: 0.0150+j0.0300 · 2-3: 0.0100+j0.0200 · 2-5: 0.0180+j0.0350 · 3-4: 0.0120+j0.0250.

Converges in 7 iterations. Final answer to check your code against:

Bus	$ V $ (p.u.)	angle (deg)	kV
1	1.00000	0.000	12.660
2	0.97477	-0.873	12.341
3	0.96626	-1.110	12.233
4	0.96342	-1.067	12.197
5	0.96908	-1.104	12.269

Branch	$ I $ (p.u.)	S_{send} (kW, kVAr)	Loss (kW)
1→2	0.87584	733.96, 477.91	11.506
2→3	0.42054	321.90, 253.82	1.769
2→5	0.17542	150.55, 81.08	0.554
3→4	0.10585	20.13, 100.28	0.134

Total loss = 13.963 kW / 27.907 kVAr. Sanity check: 733.96 kW (sent from substation) – 13.963 kW (total loss) = 720.0 kW, exactly equal to the sum of the net loads (250+300+20+150) — power balances, confirming the solution is correct.

Quick Checklist for Your Own Code (or for an AI Writing It for You)

1. Read in bus data, line data, loads, PV → convert everything to p.u.
2. Build the tree: for each bus, know its parent line and its list of child lines.
3. Flat-start all non-slack voltages at the slack voltage.
4. Loop:
 - a. **Backward:** visit buses leaves→root; compute $I_{\text{load}}(b) = \text{conj}(S(b)/V(b))$; accumulate $I(\text{parent} \rightarrow b) = I_{\text{load}}(b) + \sum \text{children's currents}$.
 - b. **Forward:** visit buses root→leaves; $V(b) = V(\text{parent}) - I(\text{parent} \rightarrow b) \cdot Z(\text{parent} \rightarrow b)$.
 - c. Check $\max |\Delta V| < \text{tolerance}$ → stop if converged.
5. Report final V per bus, I and S_{send} per branch, and losses = $|I|^2 \cdot Z$ per branch, summed for total.

Hand this document to your coding assistant (or your own code) and ask it to reproduce **both worked examples exactly** — if the numbers match, your implementation is correct for networks of any size.